MambaHSISR: Mamba Hyperspectral Image Super-Resolution

Yinghao Xu, Hao Wang, Fei Zhou, Chunbo Luo, *Senior Member, IEEE*, Xin Sun, *Senior Member, IEEE*, Susanto Rahardja, *Fellow, IEEE*, and Peng Ren, *Senior Member, IEEE*

***Abstract*— One of the main challenges facing hyperspectral image super-resolution is the complex high-dimensional data processing. Mamba leverages its ability to model long-range dependencies of linear complexity to capture the global spatial and spectral information of high-dimensional data while maintaining linear complexity. However, its visual state space equation mainly focuses on the band dimension mapping of the image, while ignoring the modeling of the spatial dimension. To overcome this limitation, we develop a Mamba hyperspectral image super-resolution framework, which comprises three essential components. The first component, i.e., the spatial Mamba subnetwork, models the spatial dimensions of hyperspectral data. It captures long-range dependencies in the pixel space, thereby integrating global spatial information into the framework. The second component, i.e., the spectral Mamba subnetwork, serves to capture long-range spectral dependencies. The third component, i.e., reconstruction, generates hyperspectral images with rich spatial and spectral details through pixel interpolation. Our Mamba framework fully develops the potential of the Mamba model in hyperspectral image super-resolution, significantly enhancing the restoration quality and accuracy of hyperspectral images. Extensive experiments on the Houston and QUST-1 datasets show that our framework outperforms state-of-the-art methods in both quantitative metrics and visual quality across diverse scenarios. We release our source code at https://gitee.com/xu_yinghao/MambaHSISR for public evaluations.**

***Index Terms*— Hyperspectral image, Mamba, spatial state, super-resolution.**

I. Introduction

HYPERSPECTRAL images contain rich spectral information, which is instrumental in accurately describing the internal structure and compositional characteristics of objects on the Earth's surface. Therefore, this type of imagery has widespread applications in various fields [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18]. Accurate predictions and analyses in these applications rely on hyperspectral images with high spatial resolution and rich spectral detail. However, the challenges in capturing high-resolution hyperspectral images arise because of the physical constraints of remote sensing satellites. To address the above issue, people utilize super-resolution techniques to enhance the spatial resolution of hyperspectral images. Hyperspectral image super-resolution employs computer vision technology to convert low spatial resolution hyperspectral images into high spatial resolution images. Therefore, hyperspectral image super-resolution is of significant importance for enhancing image quality and supporting subsequent tasks.

According to the source data, current hyperspectral image super-resolution techniques can be categorized into two primary types: one is fusion-based super-resolution [19], [20], [21] and the other is single hyperspectral image super-resolution [22], [23], [24], [25]. Fusion-based methods improve spatial resolution by combining hyperspectral images with supplementary high-resolution images, including RGB or panchromatic images. Gao et al. [26] proposed an enhanced unmixing-inspired unsupervised network with attention-embedded degradation learning. The network enhanced the representation ability of unsupervised networks through designs such as autoencoders with spatial–spectral and modality-interacted subnetworks. Yu et al. [27] proposed a generalizable unsupervised deep fusion method based on spectral–spatial collaborative constraint. The network cleverly obtained a high spatial resolution hyperspectral image through group convolution enhancement and 3-D attention factor dynamic convolutional kernels. Li et al. [28] proposed an enhanced deep image prior network. The network, built with a two-stage four-component scheme, fully utilized scene-relevant information for prior learning. It achieved high-quality hyperspectral image super-resolution performance. In contrast, single hyperspectral image super-resolution methods achieve resolution enhancement only through low-resolution hyperspectral data without the need for auxiliary information. Due to its flexibility, this method holds more advantages in practical applications. Section I-A provides a comprehensive review of single hyperspectral image super-resolution methods based on convolutional neural networks. Section I-B provides a comprehensive review of Mamba for image super-resolution.

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Yinghao Xu, Fei Zhou, and Peng Ren are with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China (e-mail: xuyinghaoupc@163.com; zhoufei@upc.edu.cn; pengren@upc.edu.cn).

Hao Wang is with the College of Control Science and Engineering, China University of Petroleum (East China), Qingdao 266580, China.

Chunbo Luo is with the Department of Computer Science, University of Exeter, EX4 4QJ Exeter, U.K. (e-mail: C.Luo@exeter.ac.uk).

Xin Sun is with the Faculty of Data Science, City University of Macau, Macau SAR, China.

Susanto Rahardja is with Singapore Institute of Technology, Singapore 138683.

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A. Convolutional Neural Networks for Hyperspectral Image Super-Resolution

Single hyperspectral image super-resolution methods have gained considerable attention due to their flexibility and practical utility. Recently, deep learning proved to be an effective approach for single-image super-resolution, primarily due to the robust feature representation and rapid inference speed of convolutional neural networks. Jiang et al. [29] proposed the spatial–spectral prior for super-resolution (SSPSR), which utilizes spatial and spectral attention residual modules to maximize spatial information and spectral correlation within high-spectral data. Group convolution and progressive upsampling are employed to address the challenge of extracting high-dimensional features from high-spectral data. Wang et al. [30] proposed the group-based embedding learning and integration network (GELIN), which reconstructs hyperspectral images group by group using a neighboring group integration module to simplify feature extraction and reconstruction. Long et al. [31] proposed a dual self-attention Swin transformer network that utilizes the ability of the shifted windows transformer in the spatial representation of both global and local features and learns spectral sequence information from adjacent bands of hyperspectral. Wang et al. [32] proposed a spectral and spatial transformer algorithm. This algorithm uses a spatial transformer structure to extract spatial features between image pixels and a spectral transformer structure to extract spectral features within the image pixels. Chen et al. [24] proposed a cross-scope spatial–spectral transformer to capture long-range dependencies in hyperspectral image super-resolution. By integrating global information, the cross-scope spatial–spectral transformer of this network enhances spatial interactions and spectral correlations. Xiao et al. [33] proposed a diffusion probabilistic model for remote sensing image super-resolution (EDiffSR). This method generates high-quality hyperspectral images through a gradual iterative process. Hu et al. [34] proposed a super-resolution in the abundance domain (SNLSR), which utilizes a spatially preserved decomposition network to estimate the abundance representation of the input hyperspectral image. The estimated low-resolution abundance is super-resolved through a spatial–spectral attention network to alleviate the challenge of incomplete modeling of high-dimensional spatial–spectral feature information. Yang et al. [35] proposed a deep blind super-resolution (DBSR) method for hyperspectral images, which integrates spatial information from blurred images to correct errors and exploits spatial and spectral information from other bands to guide feature extraction for each band. Ma et al. [36] proposed a 3D-CNN Transformer network that applies the Transformer's self-attention for global spatial interaction and uses 3D-CNN to capture spatial–spectral correlations, effectively learning priors for hyperspectral super-resolution. Li et al. [37] proposed a hybrid convolutional network (MCNet) that combines 2-D and 3-D convolutions to extract spatial and spectral features. This method fully utilizes hyperspectral information by using local feature fusion and spatial–spectral separable 3-D convolution, simplifying feature extraction in hyperspectral images. However, the high-dimensional nature of hyperspectral data makes it more difficult for convolutional neural networks to process effectively. Existing convolutional networks often demonstrate limitations when handling such high-dimensional data [38], [39], [40]. These limitations result in insufficient restoration of image details and loss of spectral information.

B. Mamba for Super-Resolution

Gu and Dao [41] first proposed the state space model (SSM). They demonstrated its effectiveness in handling long-range dependencies both mathematically and empirically. The SSM is emerging as a novel backbone in deep learning. Recently, Mamba [41], a data-dependent SSM featuring a selective mechanism and efficient hardware design, has outperformed transformers in natural language tasks and scales linearly with input length. Huang et al. [42] first integrated the state space language model into a vision backbone with linear time complexity, known as VMamba. This model employs visual state-space blocks with a 2-D selective scan across four paths, bridging the gap between 1-D scanning and 2-D visual data to capture contextual information from multiple perspectives. Mamba has widespread applications in various fields, such as classification [43], object detection [44], and super-resolution [45]. Guo et al. [46] were the first to apply the Mamba architecture to image super-resolution tasks, achieving remarkable image restoration results. Zhao et al. [47] proposed a spatial–spectral interaction super-resolution convolutional neural network–Mamba fusion network for fusion-based hyperspectral super-resolution. This network combines Mamba with convolutional layers to effectively explore both global and local features of images. Cheng et al. [45] proposed a Vision Mamba model based on a modern state space framework. This model enhances super-resolution performance by integrating MetaFormer modules and complementary attention mechanisms to activate a broader pixel area. Qiao et al. [48] proposed a hierarchical Mamba network. This network enriches spatial relationship modeling by allocating unidirectional scans of homologous and heterologous entities into cascaded Mamba groups using directionally alternating hierarchical Mamba clusters. Xiao et al. [49] designed a frequency-assisted Mamba framework that adaptively identifies perceptually critical frequency cues via an effective frequency selection module for precise super-resolution. These methods rely on external information to enhance Mamba's spatial information acquisition. However, the introduction of external information has brought about new challenges in effectively integrating these two types of information, further increasing the difficulty of reconstruction, and leading to enormous computational costs and inefficient feature extraction. Moreover, the approach mentioned above cannot fundamentally solve the incomplete spatial information extraction in the Mamba model. This challenge becomes particularly prominent when the Mamba model deals with high-dimensional hyperspectral data.

C. Challenges to Hyperspectral Image Super-Resolution

By reviewing the literature in Sections I-A and I-B, it can be observed that hyperspectral image super-resolution faces a

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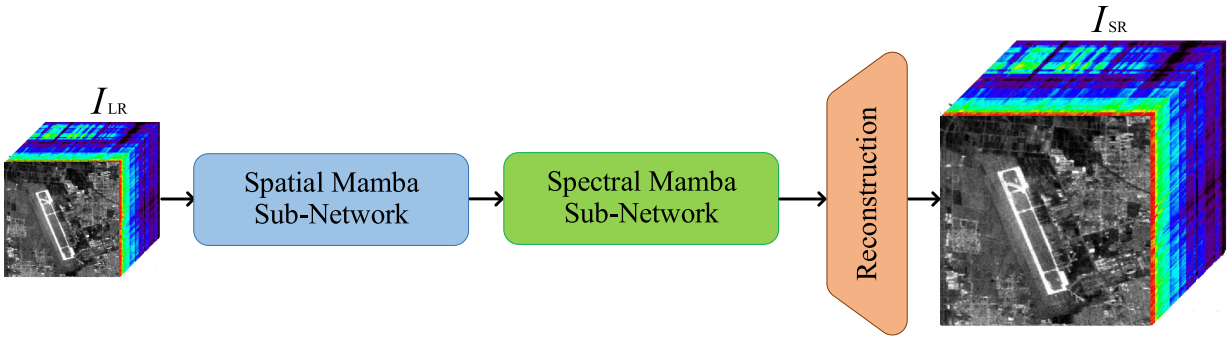


Fig. 1. Mamba hyperspectral image super-resolution.

variety of challenges. Despite the progress made in addressing certain issues in current research, several areas still require further exploration. Therefore, we have identified two specific challenges that remain central themes in the ongoing research of hyperspectral image super-resolution, as follows.

1) Limitations in exploiting the integrity of hyperspectral information. Due to the high dimensionality of hyperspectral image data, convolutional neural networks require robust feature extraction capabilities for handling complex computational tasks. However, existing convolutional neural networks exhibit limitations when processing high-dimensional data. This limitation is that the convolutional network can only explore spatial–spectral features locally, resulting in an insufficient recovery of spatial details and loss of spectral information [50]. How to exploit hyperspectral information completely, particularly to effectively maintain the integrity of spatial and spectral information for hyperspectral images, is an urgent problem that needs to be solved.
2) Limitations in the representation of spatial information. Most Mamba super-resolution models rely on external information to enhance spatial information representation. However, relying on external information not only boosts the complexity of computation but also introduces new challenges in information fusion, affecting the performance of the model. How to enable Mamba to achieve sufficient spatial information representation without relying on external information, thereby ensuring the model's strong performance in handling high-dimensional features of hyperspectral images, remains a challenging problem.

D. Contributions of Our Work

To address the challenges mentioned in Section I-C, our three new contributions to hyperspectral image super-resolution are as follows.

1) We are the first to explore the Mamba model for single hyperspectral image super-resolution. Mamba exhibits significant capability in modeling long-range dependencies with linear complexity. This capability allows it to address the challenges of insufficient detail recovery and spectral information loss in high-dimensional hyperspectral images. Our proposed framework can activate more pixels, thereby improving reconstruction quality.
2) We first proposed the Mamba hyperspectral image super-resolution framework to model long-range dependencies in pixel space. By designing the pixel spatial receptive field, we integrated rich spatial information into the model, mined hyperspectral spatial information from a global perspective, and extracted long-range spatial interaction features to promote spatial super-resolution. The model uses the spectral feature dimension as the sequence axis to capture subtle changes and patterns in global spectral information, achieving consistency in spectral reconstruction results.
3) Experimental results on the Houston and QUST-1 datasets demonstrate that the proposed diffusion framework significantly outperforms other methods in hyperspectral image super-resolution reconstruction. We have released the code for the proposed diffusion framework to encourage public evaluation (please refer to the link provided in the abstract).

II. Mamba Hyperspectral Image Super-Resolution Framework

As illustrated in Fig. 1, our Mamba hyperspectral image super-resolution framework consists of three components: spatial feature extraction, spectral feature extraction, and high-quality reconstruction. Our framework effectively handles large spatial-scale high-dimensional hyperspectral images by leveraging the advantage of modeling remote dependencies under linear complexity. First, in the spatial feature extraction component, we proposed the spatial Mamba subnetwork, which uses 2-D spatial features as the sequence axis to model the long-distance dependencies between pixels. This process generates spatial information containing detailed features of global objects. Second, in the spectral feature extraction component, we designed the spectral Mamba subnetwork, which uses the spectral feature dimension as the sequence axis to capture the global spectral information and patterns with subtle changes in the extracted spatial information, calibrate the captured spectral features, and finally generate an optimized spectral representation. Finally, in the high-quality reconstruction component, the optimized spectral representation undergoes pixel interpolation for upsampling, resulting in a hyperspectral image with abundant spatial and spectral details.

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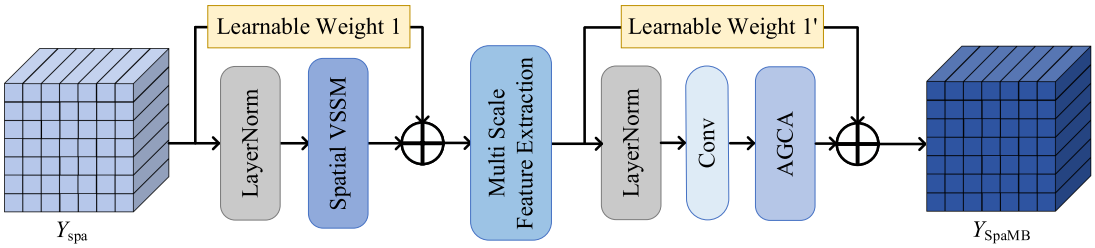


Fig. 2. Spatial Mamba block.

III. Spatial Mamba Subnetwork for Utilizing Spatial Information

In hyperspectral image super-resolution, it is crucial to fully utilize spatial information; otherwise, it will result in poor image detail recovery. We found that existing Mamba-based image super-resolution methods often directly use the Mamba model for mapping image band sequences during spatial information processing. This approach cannot fully extract the spatial information of the image and greatly affects the restoration of spatial details. Therefore, we designed the spatial Mamba subnetwork to achieve efficient use of spatial information. The subnetwork consists of multiple spatial Mamba blocks, each of which can efficiently extract the spatial features of hyperspectral images. Our spatial Mamba block primarily consists of three components: a vision spatial state-space module (SpaVSSM), a multiscale feature extraction module, and an adaptive graph channel attention (AGCA) mechanism [51]. The structure of the spatial Mamba block is shown in Fig. 2.

For the input image $I_{LR} \in R^{H\times W\times C}$, we first transform it into spatial features via convolution

$$Y_{spa} = \text{Conv}(I_{LR}) \tag{1}$$

where Conv(·) represents the convolution operation, and Y_{spa} represents the spatial features, which carry all the spatial information of the image.

The SpaVSSM enables anchor pixels to perceive a wider range of neighborhoods through spatial dimension mapping, capturing a wider range of pixel spatial long-range dependencies. Specifically, the input feature tensor $Y_{spa}\in R^{H\times W\times C'}$ is projected into a higher dimensional space through a linear projection layer, resulting in the feature tensor $YZ\in R^{H\times W\times C''}$. Subsequently, YZ is split along the final dimension into spatial characteristics Y'_{spa} and activation feature Z, each with an augmented channel count C'. The spatial feature Y''_{spa} is converted into a nonlinear output through the SiLU activation function after convolution, and then a spatial transformation operation is performed to generate the core spatial feature Y''_{spa}. Later, Y''_{spa} is restored to its original $H \times W$ size through a reshape operation. The reshaped Y'_{spa} is combined with the activation characteristics Z via the Hadamard product to integrate various features. Ultimately, the various features are projected back to the shape of $H \times W \times C'$, generating the final output Y_{Spaout}

$$\{Y'_{Spa}, Z\} = \text{Split}\big(\text{Linear}(Y_{spa})\big), \quad Y'_{spa} \in R^{HW\times C'} \quad Z \in R^{HW\times C'} \tag{2}$$

$$rY''_{Spa} = \text{Reshape}\Big(\text{Spatialtran}\Big(\text{SiLU}\Big(\text{Conv}(Y'_{spa})\Big)\Big)\Big) \quad Y''_{spa} \in R^{H\times W} \tag{3}$$

$$Y_{Spaout} = \text{Linear}\ (Y''_{Spa} \odot Z), \quad Y_{Spaout} \in R^{H\times W\times C} \tag{4}$$

where linear(·) represents the linear projection layer, Split(·) represents the splitting operation, and Spatialtran(·) represents the spatial transformation operation, which includes the squeeze and expand operations of spatial transformation. Reshape(·) represents the reshaping of spatial features, and ⊙ represents the Hadamard product operation. Specifically, the data transformation process within this SpaVSSM is illustrated in Fig. 3. SpaVSSM employs a four-direction scan of spatial information, with the first direction explained as an example. First, the spatial dimension of the spatial information is mapped into a higher dimensional space. Then, Spatialtran Squeeze is applied to extract the core spatial features (A, B, C, and D). In A, B, C, and D, the core spatial features extracted from all spatial information $(1, 2, 3, 4, \ldots, 8, 9)$ are preserved. Finally, by reshaping the core spatial features through Spatialtran Expend, the spatial features $(1', 2', 3', 4', \ldots, 8', 9')$ are obtained, which retain the complete spatial structure and details.

In addition, we introduce a learnable weight factor, LW1, to control the fusion of information at different levels. The above process can be expressed by the following:

$$Y_{SpaVSSMout} = Y_{Spaout} + \text{LW1} \cdot Y_{spa}. \tag{5}$$

For the multiscale feature extraction module, extracting multiscale features from the 2-D spatial domain restores the fragmentation of local spatial information caused by SpaVSSM and improves the structures and representation of spatial information. The spatial features with complete local spatial structures and long-range pixel dependencies are obtained through the multiscale feature extraction module as follows:

$$\begin{aligned} Y_{MSFE} = {} & \text{ReLU}\big(\text{BN}\big(\text{Conv1}\times 1(Y_{SpaVSSMout})\big)\big) \\ & + \text{ReLU}\big(\text{BN}\big(\text{Conv3}\times 3(Y_{SpaVSSMout})\big)\big) \\ & + \text{ReLU}\big(\text{BN}\big(\text{Conv5}\times 5(Y_{SpaVSSMout})\big)\big). \end{aligned} \tag{6}$$

The AGCA enables spatial Mamba blocks to focus on learning diverse spatial feature representations. It also selects key spatial features, mitigating spatial feature redundancy. The process outlined above can be expressed as

$$\begin{aligned} Y_{SpaMB} = {} & \text{AGCA}(\text{Conv}(\text{LayerNorm}(Y_{MSFE}))) \\ & + \text{LW1}' \cdot Y_{MSFE} \end{aligned} \tag{7}$$

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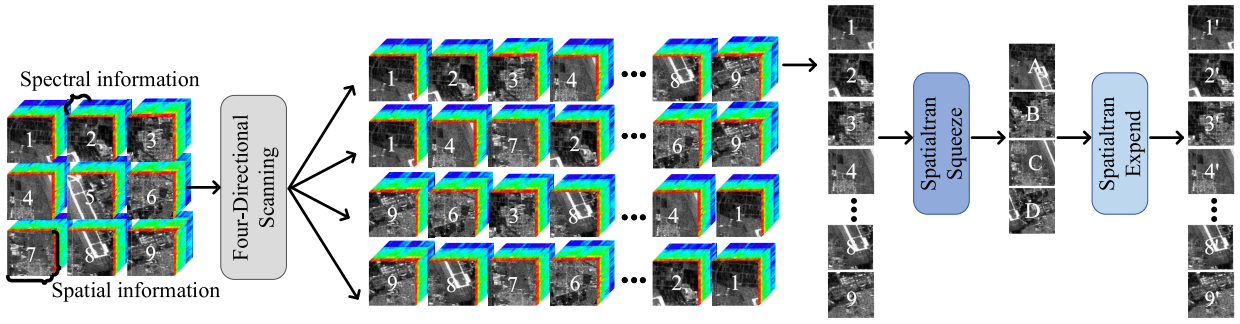


Fig. 3. Data changes representing the spatial vision SSM operation in the spatial Mamba block.

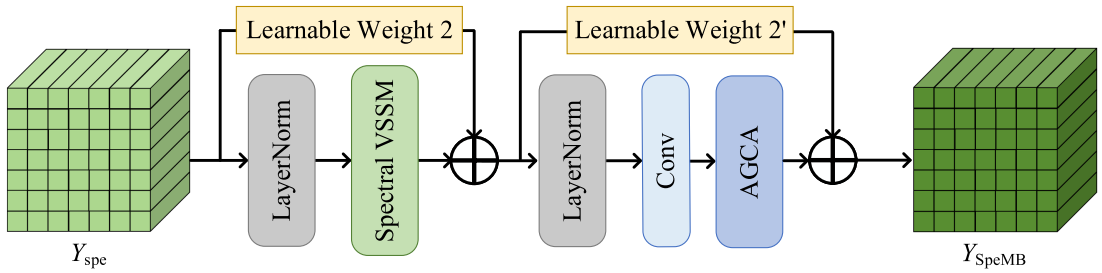


Fig. 4. Spectral Mamba block.

where Y_{SpaMB} represents the key spatial feature representation, AGCA(·) represents the adaptive graph channel attention mechanism operation, and LW1′ represents the current learnable weight factor.

Here, the spatial Mamba subnetwork effectively captures long-range spatial dependencies in hyperspectral images while preserving local spatial structure integrity. It mitigates feature redundancy and enhances representation, enabling the model to effectively capture diverse spatial features. This is crucial for significantly enhancing spatial detail restoration in Mamba hyperspectral image super-resolution.

IV. Spectral Mamba Subnetwork for Utilizing Spectral Information

In traditional hyperspectral image super-resolution methods, attention mechanisms are usually utilized to capture valuable contextual information. However, these methods only explore spectral features within the local spectral receptive field and it is difficult to fully mine the rich global information in hyperspectral images. To address this issue, we designed the spectral Mamba subnetwork with spectral feature dimensions as the sequence axis. This subnetwork ensures the accuracy of spectral reconstruction by capturing subtle changes and patterns in global spectral information. The spectral Mamba subnetwork consists of M spectral Mamba blocks. Our spectral Mamba block primarily consists of two components: a vision spectral state-space module (SpeVSSM) and an AGCA, as shown in Fig. 4.

For the SpeVSSM, its spectral feature dimensions serve as the sequence axis, capturing a wider range of long-range spectral dependencies. The spectral information primarily consists of 1-D features. Therefore, bidirectional scanning (from the first band to the final band) is sufficient to capture relevant information along the spectral dimension. Specifically, the input the feature tensor Y_{spe}, which is mapped to a higher dimensional space through a linear projection layer to generate a feature tensor $Y'Z' \in R^{HW \times C''}$. Here, Y_{spe} is obtained by convolving Y_{SpaMB} with a convolutional layer. Then, $Y'Z'$ is divided along the first dimension into spectral features Y'_{spe} and activation features Z, both of which have the same number of channels as HW. The spectral characteristics Y'_{spe} undergo convolution and a SiLU activation function to include nonlinearity, followed by a spectral transformation operation to provide core spectral features Y''_{spe}. The Y''_{spe} tensor is then reconfigured to its original dimensions of C''. Subsequently, Y''_{spe} is combined with the activation characteristics Z via the Hadamard product to integrate many features. Ultimately, the integration of many features is reverted to the dimensions $H \times W \times C''$, yielding the final output Y_{Speout}

$$\{Y'_{Spe}, Z'\} = \text{Split}\big(\text{Linear}(Y_{spe})\big), \quad Y'_{Spe} \in R^{HW \times C'}, \quad Z' \in R^{HW \times C'} \tag{8}$$

$$Y''_{Spe} = \text{Reshape}\big(\text{Spectraltran}\big(\text{SiLU}\big(\text{Conv}(Y_{Spe})\big)\big)\big), \quad Y''_{Spe} \in R^{C''} \tag{9}$$

$$Y_{Speout} = \text{Linear}\left(Y''_{Spe} \odot Z'\right), \quad Y_{Speout} \in R^{H \times W \times C'}. \tag{10}$$

Specifically, the transformation process of hyperspectral image data within the vision spectral state-space module is illustrated in Fig. 5. SpeVSSM employs a two-directional scan of spectral information to generate a scanning sequence. Taking the first scanning sequence as an example, the spectral dimension of the spectral information is first serialized, and

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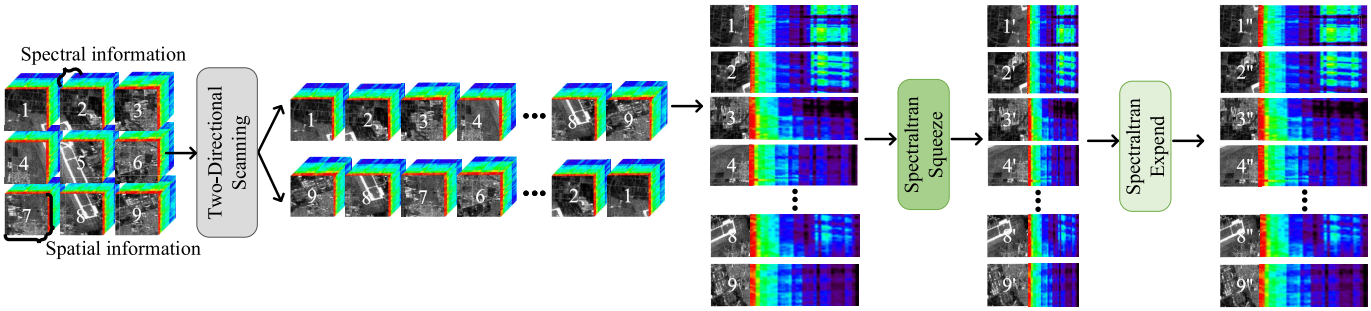


Fig. 5. Data changes representing the SpeVSSM operation in the spectral Mamba block.

then, through Spectraltran Squeeze, the core spectral features (with a reduced spectral dimension) are extracted. The core spectral features $(1, 2, 3, 4, \ldots, 8, 9)$ retain the essential spectral characteristics extracted from all spectral information $(1', 2', 3', 4', \ldots, 8', 9')$. Finally, after reshaping the core spectral features through Spectraltran Expend, the spectral features $(1'', 2'', 3'', 4'', \ldots, 8'', 9'')$ are obtained, which encapsulate the complete spectral structure and details. To further control hierarchical information, we introduce the learnable weight factor LW2. The process outlined above can be defined as follows:

$$Y_{\text{SpeVSSMout}} = Y_{\text{Speout}} + \text{LW2} \cdot Y_{\text{speMB}}. \tag{11}$$

For the AGCA, it effectively measures spectral similarity and focuses on learning diverse spectral feature representations. This allows AGCA to select and highlight key spectral features, thereby improving spectral information utilization efficiency. The above process can be expressed as

$$Y_{\text{SpeMB}} = \text{AGCA}\big(\text{Conv}\big(\text{LayerNorm}(Y_{\text{SpeVSSMout}})\big)\big) + \text{LW2}' \cdot Y_{\text{SpeVSSMout}}. \tag{12}$$

Here, the spectral Mamba subnetwork effectively captures spectral self-similarity and selects key spectral features by modeling long-range dependencies on the spectrum, thus achieving spectral consistency results.

V. High-Quality Reconstruction for Spatial–Spectral Information Fusion

Our Mamba hyperspectral image super-resolution framework utilizes a progressive upsampling method, incrementally enhancing image details over many iterative upsampling components. This method mitigates the complexity and challenges of high-dimensional data reconstruction by segmenting the super-resolution reconstruction process into several steps. The formulae for upsampling and reconstruction are as follows:

$$I_{\text{SR}} = \text{Conv}\big(\text{PixelShuffle}\big(\text{Conv}\big(\text{PixelShuffle}\big(\text{LeakyReLU}\big(\text{Conv}\big(Y_{\text{SpeMB}}\big)\big)\big)\big)\big)\big) \tag{13}$$

where LeakyReLU$(\cdot)$ represents the activation function and PixelShuffle$(\cdot)$ represents the upsampling method. In the Mamba hyperspectral image super-resolution framework, a progressive upsampling method is used to refine image details incrementally, simplifying the reconstruction process and enhancing the quality of high-dimensional data reconstruction.

The reconstruction process is simplified through progressive upsampling. By upsampling in multiple stages, this strategy reduces the upsampling gap at each step. In contrast, a single-step upsampling method lacks the ability to refine features gradually, which may hinder the generation of fine details. Each stage of progressive upsampling focuses on features at a specific scale, ensuring a more stable and precise reconstruction process.

For Mamba hyperspectral image super-resolution training, we input low-resolution images, which undergo spatial feature extraction, spectral feature extraction, and high-quality reconstruction to produce super-resolution images. During the training process, we use the $L1$ loss as the loss function, encouraging the framework to generate high-fidelity images.

VI. Experimental Results and Discussions

A. Dataset

To evaluate the effectiveness of our proposed framework, experiments are conducted on real data from multiple scenarios. The dataset, as illustrated in Fig. 6, consists of two hyperspectral images. These images cover a range of spatial resolutions from 1 to 10 m and include typical targets.

The Houston dataset,[1] obtained on February 16, 2017, comprises hyperspectral imagery of the Houston University campus and adjacent metropolitan regions. It has a spatial resolution of 1 m and comprises 48 bands ranging from 380 to 1050 nm. The dimensions of the photographs are 4172 by 1202 pixels.

The QUST-1 satellite dataset,[2] acquired on April 10, 2019, includes hyperspectral images of the Qingdao region in Shandong, China, encompassing various geographical features, including plains, mountains, and oceans. The spatial resolution is 10 m, encompassing 32 bands ranging from 400 to 1000 nm. The dimensions of each image are 5056 by 5056 pixels.

B. Evaluation Metrics

This study employs eight popular evaluation metrics to comprehensively assess the performance of various methods

[1]https://hyperspectral.ee.uh.edu/?page_id=1075
[2]https://www.obtdata.com/#/dataExpress

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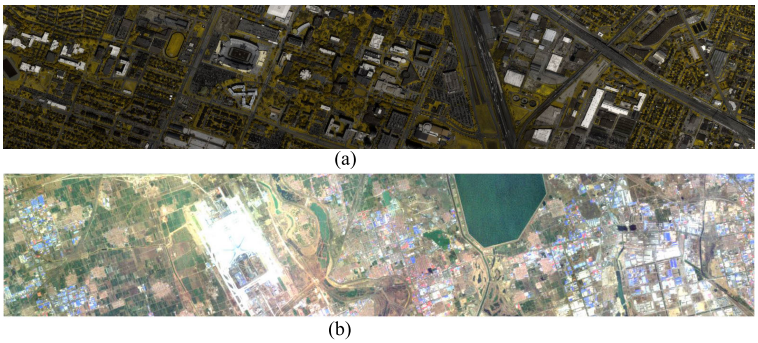


Fig. 6. Experimental data including multiple and diverse scenes. (a) Houston. (b) QUST-1 Qingdao, China.

in terms of spatial, spectral, and visual perception aspects. The metrics encompass Fréchet inception distance (FID) [52], peak signal-to-noise ratio (PSNR) [53], structural similarity index (SSIM) [54], spectral angle mapper (SAM) [55], global dimensionless relative error of synthesis (ERGAS) [56], correlation coefficient (CC) [57], learned perceptual image patch similarity (LPIPS) [58], and deep image structure and texture similarity (DISTS) [59]. FID is widely used to assess the generative quality of models, combining mean and covariance to more accurately represent distribution differences in high-dimensional spaces. PSNR evaluates the quality of image super-resolution by analyzing the discrepancies in pixel values between the generated and original images. Nevertheless, it fails to consider the human eye's sensitivity to specific details and structures. SSIM evaluates image similarity by analyzing pixel value differences as well as brightness, contrast, and structure to more accurately represent human perception. SAM assesses spectral similarity by determining the angle between spectral vectors, hence evaluating the coherence of spectral data. ERGAS quantifies the global relative error between reconstructed and reference images. CC quantifies the linear correlation between two variables, indicating the consistency between produced and original images in the spectral domain. LPIPS evaluates perceptual similarity by simulating human visual perception using deep learning models while considering the sensitivity of the human eye to image features and structures. DISTS amalgamates structural and textural similarity, delivering a thorough evaluation of visual resemblance across images. An increase in PSNR, SSIM, and CC values indicates excellent performance and a decrease in SAM, ERGAS, FID, LPIPS, and DISTS values signifies an enhancement in super-resolution quality.

C. Implementation Details

We divide each dataset into training data and test data, 80% for training and 20% for testing. When dividing the training dataset, we adopted a patch-based approach, which extracts overlapping regions from high-resolution hyperspectral images as training datasets. When the scale factor is 2, the extracted patches are 64 × 64 pixels with an overlap of 16 pixels. When the scale factor is 3, the extracted patches are 87 × 87 pixels with an overlap of 84 pixels. When the scale factor is 4, the patch size is still 64 × 64 pixels, but the overlap increases to 32 pixels. Then a degradation model is introduced, which generates a training set by simulating the blurring effect on remote sensing images using different disk blur kernels. Finally, the photographs are downsampled using bicubic interpolation to generate low-resolution images with different levels of blur.

Our proposed Mamba hyperspectral image super-resolution framework is implemented using PyTorch. All tests are performed on a machine with two NVIDIA GeForce RTX A6000 GPUs, each possessing 48 GB of RAM and an AMD EPYC 7742 CPU. In our Mamba hyperspectral image super-resolution framework, the number of spatial Mamba blocks and spectral Mamba blocks are set to 15 and 3, respectively. We use the Adam optimizer in conjunction with the ReduceLROnPlateau learning rate scheduler. Throughout the training process, we monitor the performance parameters of the PSNR. We set the initial learning rate to 0.0004. If there is no improvement in the initialization distance over two consecutive epochs, the learning rate is automatically reduced by 0.25 times to enhance subsequent training performance. Considering the network's magnitude, we randomly choose 96 low-resolution images from the training dataset for each training batch. At the same time, to improve the generalization ability of the framework, we randomly rotated and flipped the data.

D. Performance Evaluation

Our hyperspectral Mamba super-resolution framework is assessed against seven leading hyperspectral image super-resolution methods at × 2, × 3, and × 4 upsampling factors. The techniques include Bicubic, MCNet [37], SSPSR [29], GELIN [30], EDiffSR [33], SNLSR [34], and DBSR [35]. Bicubic functions as a conventional benchmark for interpolation. MCNet, SSPSR, GELIN, SNLSR, and DBSR are approaches based on convolution. MCNet, using hybrid 2-D–3-D convolution, exemplifies a cutting-edge methodology. SSPSR, using residual blocks, is likewise cutting-edge. GELIN integrates group embedding learning with the network. EDiffSR distinguishes itself for its advanced performance in remote sensing imagery using a diffusion model. SNLSR tackles super-resolution by transitioning the issue to the abundance domain, while DBSR amalgamates spatial information from blurred images to rectify inaccuracies. We use publicly accessible PyTorch implementations of these methodologies to guarantee an equitable comparison. All techniques are

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TABLE I
QUANTITATIVE RESULTS OF HYPERSPECTRAL IMAGE SUPER-RESOLUTION ON THE HOUSTON DATASET

Scale	Metric	Bicubic	MCNet [37]	SSPSR [29]	GELIN [30]	EDiffSR [33]	SNLSR [34]	DBSR [35]	Our
×2	FID ↓	133.9400	120.9811	102.3420	71.3090	62.3189	57.7544	56.8473	**55.2556**
	PSNR ↑	35.4262	38.2980	38.9625	40.0926	40.5046	40.1850	40.7810	**40.9672**
	SSIM ↑	0.9334	0.9276	0.9275	0.9783	**0.9807**	0.9802	0.9786	0.9803
	SAM ↓	3.5435	2.1700	2.1089	2.3091	2.5247	1.8384	2.0373	**1.8006**
	ERGAS ↓	1.9368	1.5530	1.8178	1.5458	1.5142	1.2376	1.4636	**1.2029**
	CC ↑	0.9503	0.9757	0.9759	0.9868	0.9872	0.9962	0.9936	**0.9940**
	LPIPS ↓	0.0748	0.0544	0.0497	0.0317	0.0276	0.0248	0.0240	**0.0198**
	DISTS ↓	0.2080	0.1718	0.1654	0.1491	0.1347	0.1640	0.1487	**0.1433**
×3	FID ↓	150.2151	132.2624	130.5351	104.3374	95.0853	90.3573	86.6155	**83.6988**
	PSNR ↑	33.0243	35.2905	35.8343	37.0786	37.5167	37.4341	**38.0237**	38.0164
	SSIM ↑	0.9172	0.9213	0.9238	0.9544	0.9495	0.9546	0.9541	**0.9553**
	SAM ↓	3.7287	2.5513	2.5112	2.3131	2.6651	1.9714	2.1317	**1.8691**
	ERGAS ↓	2.3480	1.6700	1.7726	1.6490	1.6005	1.5904	1.6673	**1.4801**
	CC ↑	0.9213	0.9394	0.9398	0.9413	0.9457	0.9621	0.9622	**0.9624**
	LPIPS ↓	0.1057	0.0825	0.0797	0.0616	0.0603	0.0490	0.0516	**0.0410**
	DISTS ↓	0.2210	0.1638	0.1758	0.1383	0.1262	0.1403	0.1344	**0.1288**
×4	FID ↓	172.6825	149.0857	164.1773	141.7199	131.8260	126.7590	120.0227	**115.6536**
	PSNR ↑	32.5335	34.3566	34.7986	36.2364	36.7218	36.8713	37.4788	**37.6860**
	SSIM ↑	0.9037	0.9178	0.9229	0.9335	0.9213	0.9318	0.9325	**0.9334**
	SAM ↓	4.0193	3.0048	2.9844	2.3826	2.8815	2.1603	2.2865	**1.9909**
	ERGAS ↓	2.8566	1.8568	1.8010	1.8207	**1.7535**	2.0096	1.9404	1.8195
	CC ↑	0.9457	0.9581	0.9587	0.9507	0.9600	**0.9842**	0.9875	0.9870
	LPIPS ↓	0.1370	0.1109	0.1099	0.0917	0.0931	0.0733	0.0794	**0.0624**
	DISTS ↓	0.2404	0.1606	0.1913	0.1315	0.1214	0.1206	0.1240	**0.1180**

trained on an identical dataset using parameters specified in their papers to ensure the fairness and trustworthiness of the findings.

1) Experimental Results on the Houston Dataset:

a) Quantitative results: Table I summarized the quantitative super-resolution performance comparison results of our Mamba framework and seven other methods on the Houston dataset. The optimal outcomes are emphasized in bold, whereas the suboptimal results are underscored. Our Mamba framework surpasses most other methods across practically all criteria at scale factors of 2, 3, and 4. Regarding FID, our Mamba hyperspectral image super-resolution framework outperforms the second-best method DBSR by reducing 1.5917, 2.9167, and 4.3691 points at scale factors 2, 3, and 4, respectively, demonstrating higher visual fidelity. For PSNR, our framework beats DBSR by 0.1862 dB at scale factor 2 and by 0.2072 dB at scale factor 4, exhibiting greater reconstruction quality. For SSIM, our framework is 0.0007 lower than DBSR at scale factor 3 and 0.0003 lower at scale factor 4, respectively. However, our framework still displays outstanding overall performance. For SAM, our approach decreases the score by 0.0378, 0.1023, and 0.1694 relative to SNLSR at scale factors 2, 3, and 4, respectively, indicating enhanced spectral integrity. The ERGAS metric illustrates that our Mamba framework surpasses the second-best method by 0.0347 at a scale factor of 2. Our framework attains a peak CC metric value of 0.9940 at a scale factor of 2, demonstrating a robust correlation with the ground-truth data. Finally, for the perceptual metrics LPIPS and DISTS, our framework attains optimal scores of 0.0198 and 0.1433 at a scale factor of 2, 0.0410 and 0.1228 at a scale factor of 3, and 0.0624 and 0.1180 at a scale factor of 4, corroborating its efficacy in generating high-quality images. Our Mamba framework effectively exploits extensive long-range pixel dependencies and captures subtle variations in global spectral information, excelling in both spatial and spectral feature utilization. This is achieved through the integration of the spatial Mamba subnetwork, spectral Mamba subnetwork, and high-quality reconstruction, yielding high-resolution image outputs.

b) Visual effects: Figs. 7 and 8 illustrate the spatial super-resolution outcomes of the spectral bands 29–26–19 as RGB from the experimental region at scale factors of 2, 3, and 4, which offer a qualitative comparison of various methodologies. To better visualize, the same areas marked with red rectangles are enlarged. The results demonstrate how well our Mamba hyperspectral super-resolution framework restores the spatial structure of hyperspectral pictures, while other methods often produce outputs that are too smooth or display aberrations like mosaics. Our Mamba framework improves the effectiveness of hyperspectral image use over current techniques, especially by maintaining the integrity of both spatial and spectral information. Furthermore, to restore the integrity of spatial structures, We modified the spatial Mamba subnetwork to capture large long-range pixel relationships, which yield more accurate information. As deep learning-based methods, MCNet, SSPSR, and GELIN recreate certain texture details but have trouble making the most of spatial data. As shown in the target on the left side of the red rectangle, this results in an inadequate reconstruction of fine edge contours. EDiffSR, as a diffusion-based method, restores spatial details but overlooks the long-range dependencies in the spectral dimension, leading to subpar performance in multiband remote sensing hyperspectral images. The SNLSR method tackles the super-resolution issue by transitioning to the abundance domain, but it fails to enhance the receptive field in the pixel space, leading to an incomplete reconstruction of global spatial structures. DBSR mitigates reconstruction challenges via group-based super-resolution, resulting in enhanced spatial

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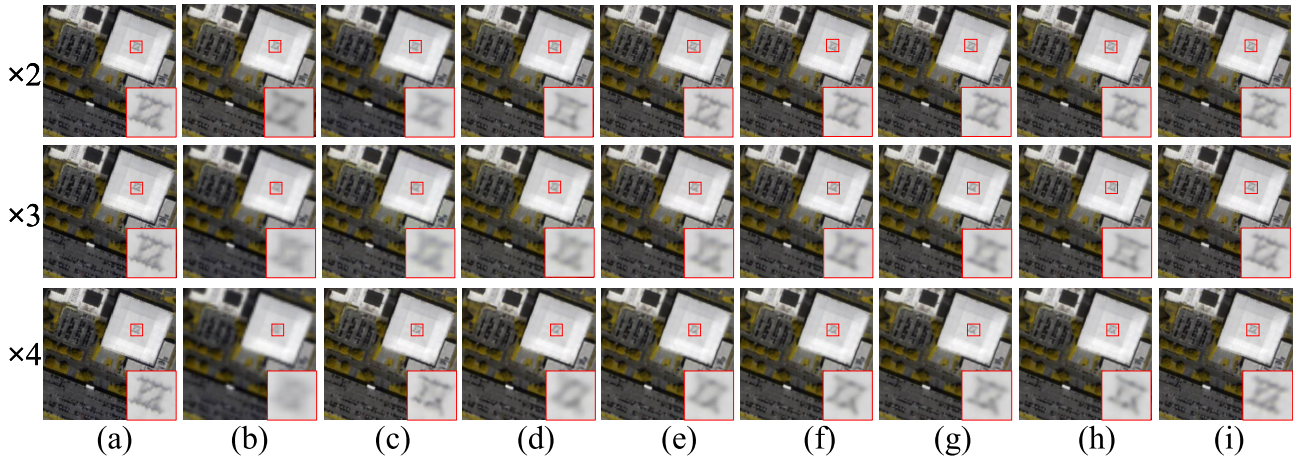


Fig. 7. Quantitative results of hyperspectral image super-resolution on the Houston dataset. (a) Ground truth. (b) Bicubic. (c) MCNet [37]. (d) SSPSR [29]. (e) GELIN [30]. (f) EDiffSR [33]. (g) SNLSR [34]. (h) DBSR [35]. (i) Our framework.

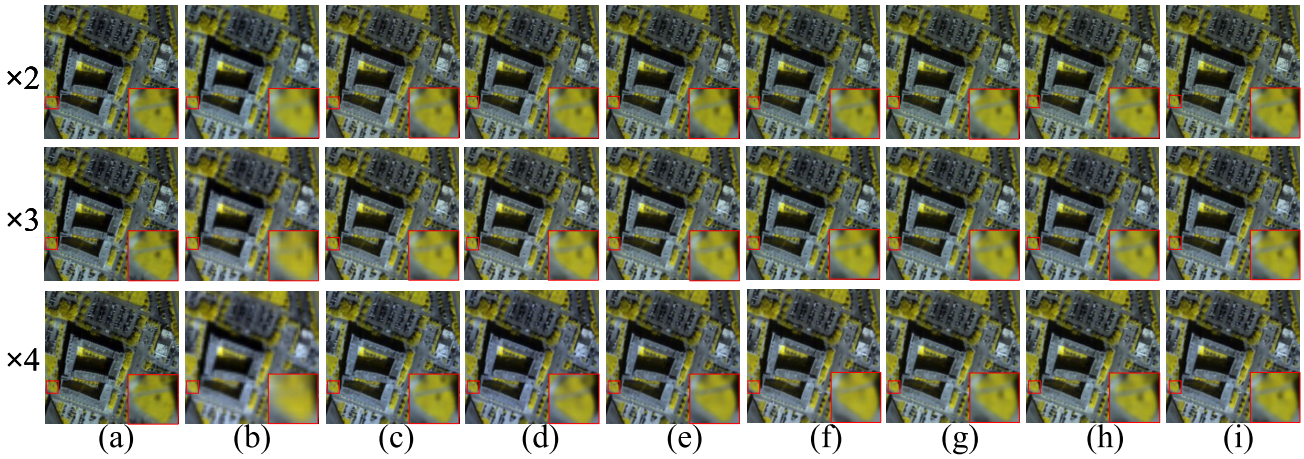


Fig. 8. Quantitative results of hyperspectral image super-resolution on the Houston dataset. (a) Ground truth. (b) Bicubic. (c) MCNet [37]. (d) SSPSR [29]. (e) GELIN [30]. (f) EDiffSR [33]. (g) SNLSR [34]. (h) DBSR [35]. (i) Our framework.

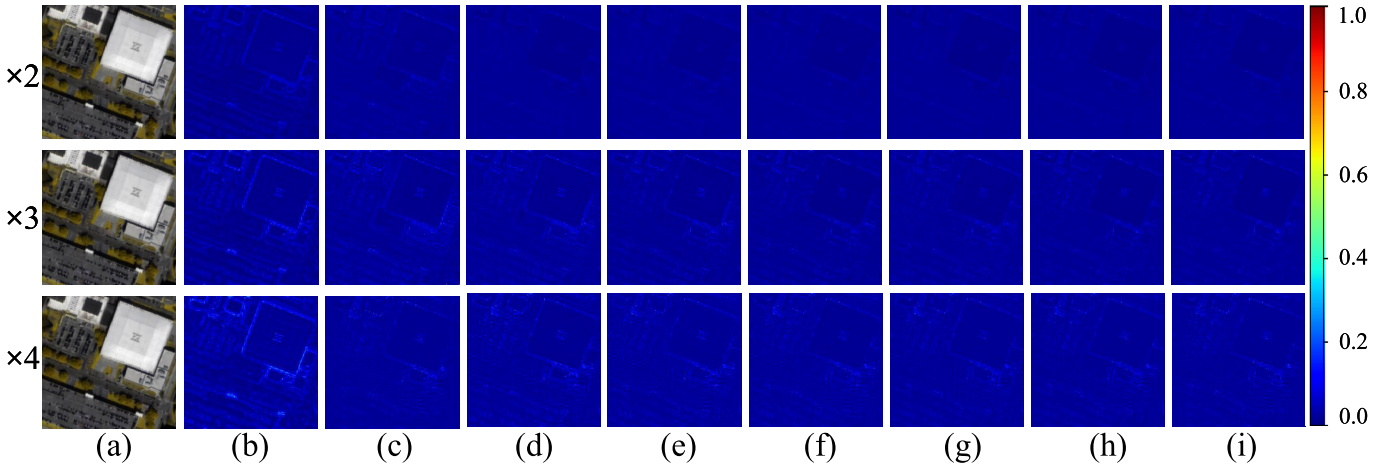


Fig. 9. Quantitative results of hyperspectral image super-resolution on the Houston dataset. (a) Ground truth. (b) Bicubic. (c) MCNet [37]. (d) SSPSR [29]. (e) GELIN [30]. (f) EDiffSR [33]. (g) SNLSR [34]. (h) DBSR [35]. (i) Our framework.

reconstruction performance. However, it does not fully utilize spectral information beyond the groups. In contrast, our Mamba hyperspectral super-resolution framework enhances the pixel space receptive field by modeling long-range dependencies within the pixel space. This allows the framework to access information from a wider domain, enhancing spatial super-resolution and yielding more realistic and sharper visual outcomes.

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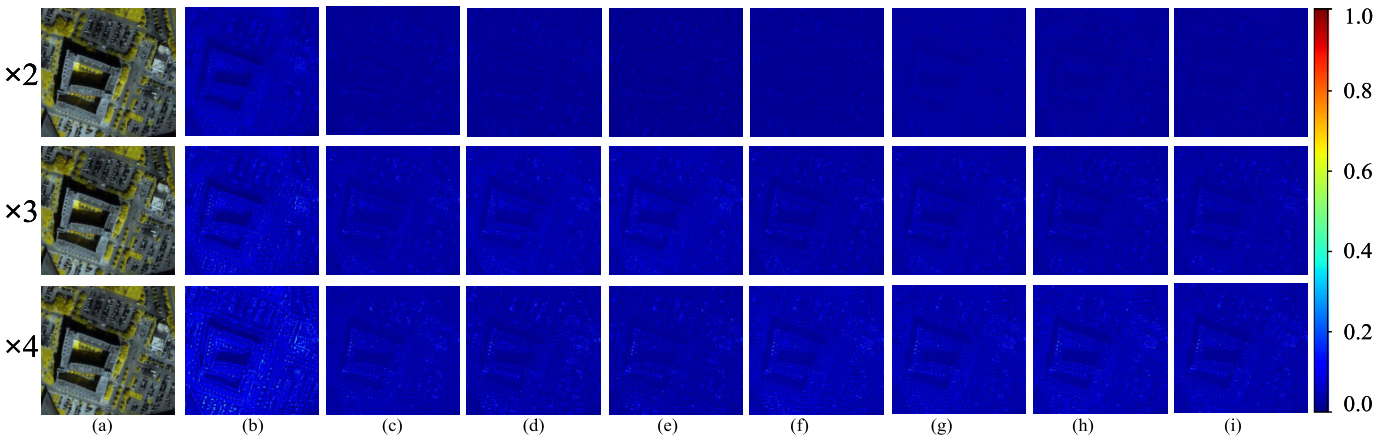


Fig. 10. Quantitative results of hyperspectral image super-resolution on the Houston dataset. (a) Ground truth. (b) Bicubic. (c) MCNet [37]. (d) SSPSR [29]. (e) GELIN [30]. (f) EDiffSR [33]. (g) SNLSR [34]. (h) DBSR [35]. (i) Our framework.

TABLE II
QUANTITATIVE RESULTS OF HYPERSPECTRAL IMAGE SUPER-RESOLUTION ON THE QUST-1 DATASET

Scale	Metric	Bicubic	MCNet [37]	SSPSR [29]	GELIN [30]	EDiffSR [33]	SNLSR [34]	DBSR [35]	Our
×2	FID ↓	147.4573	128.6295	117.4573	97.4800	74.1788	68.6293	60.2814	**53.4462**
	PSNR ↑	36.2397	38.6229	39.9415	40.7533	41.2151	41.6543	41.9626	**42.2770**
	SSIM ↑	0.9168	0.9284	0.9303	0.9811	0.9627	0.9587	**0.9838**	0.9797
	SAM ↓	5.3414	3.2103	4.0193	3.3022	3.8842	**2.3417**	2.2889	2.3641
	ERGAS ↓	2.1508	1.8858	2.1730	1.5055	1.4800	1.7763	1.5823	**1.2749**
	CC ↑	0.9310	0.9731	0.9769	0.9808	0.9818	0.9693	0.9842	**0.9880**
	LPIPS ↓	0.1157	0.0644	0.0579	0.0402	0.0387	0.0366	0.0363	**0.0365**
	DISTS ↓	0.2775	0.1492	0.1300	0.0769	0.1654	0.0603	0.0585	**0.0498**
×3	FID ↓	150.5202	130.9029	115.6517	99.8469	77.5448	71.7173	65.7948	**61.1523**
	PSNR ↑	33.6994	36.2074	37.0163	37.7158	38.0598	38.3701	38.5392	**38.6330**
	SSIM ↑	0.8696	0.8920	0.9008	0.9363	0.9323	0.9155	**0.9481**	**0.9475**
	SAM ↓	6.9453	4.4897	4.3456	3.7882	4.2366	3.4411	2.8604	**2.8210**
	ERGAS ↓	3.0576	2.7011	2.4698	1.7773	1.6155	**1.4589**	1.5896	1.4855
	CC ↑	0.8727	0.9039	0.9025	0.9170	0.9196	0.8968	0.9160	**0.9240**
	LPIPS ↓	0.1439	0.0857	0.0781	0.0483	0.0334	0.0333	0.0362	**0.0315**
	DISTS ↓	0.2698	0.1778	0.1524	0.1164	0.1455	0.0980	0.0929	**0.0878**
×4	FID ↓	176.1066	138.5962	122.7125	108.3749	84.1157	77.7684	74.0261	**71.3842**
	PSNR ↑	31.0788	35.9233	36.2697	36.8855	37.1293	37.3465	37.5816	**37.8659**
	SSIM ↑	0.8250	0.8583	0.8742	0.8942	0.9047	0.8751	0.9152	**0.9182**
	SAM ↓	8.7479	5.8975	4.7962	4.3821	4.7107	4.6392	3.5123	**3.3591**
	ERGAS ↓	4.0924	3.6293	2.8693	1.4722	1.7651	1.8524	**1.7171**	1.7580
	CC ↑	0.8652	0.8874	0.8807	0.9072	0.9112	0.8772	0.9015	**0.9136**
	LPIPS ↓	0.1725	0.1073	0.0986	0.0566	0.0283	0.0303	0.0362	**0.0266**
	DISTS ↓	0.2698	0.2115	0.1792	0.1593	0.1297	0.1385	0.1299	**0.1283**

Figs. 9 and 10 present the absolute error maps of the spectral bands 29–26–19 as RGB across various methods at scale factors of 2, 3, and 4. These error maps highlight the differences between the reconstructed images and the ground truth, with smaller error maps indicating better performance of the super-resolution methods. Figs. 9 and 10 illustrate that the images generated by our Mamba framework demonstrate minimal error. The reconstructed images produced by our Mamba framework preserve spatial information that closely corresponds to the ground truth. Our Mamba framework demonstrates superior performance in detail and texture reconstruction, due to the effective integration of its three core components. This integration maximizes the utilization of hyperspectral information, leading to enhanced spatial fidelity in the reconstructed images.

Fig. 15(a) illustrates the spectral profiles of selected pixels (highlighted as red dots) in the reconstructed images from the Houston dataset at a scale factor of 4. The spectral curves produced by our Mamba hyperspectral super-resolution method exhibit a stronger similarity to the ground truth compared to those from other techniques, highlighting the Mamba model's capability to effectively retain spectral features.

2) Experimental Results on QUST-1 Dataset:

a) Quantitative results: Table II presents the quantitative super-resolution performance of seven methods on the QUST-1 Qingdao region dataset. The top results are highlighted in bold, and the second-best results are underlined. Our proposed Mamba hyperspectral super-resolution framework outperforms most state-of-the-art methods on the majority of metrics at scale factors 2, 3, and 4. These results indicate that our Mamba framework is highly effective for hyperspectral super-resolution tasks, demonstrating superior performance.

b) Visual effects: Figs. 11 and 12 illustrate the visual results of various super-resolution methods on the QUST-1

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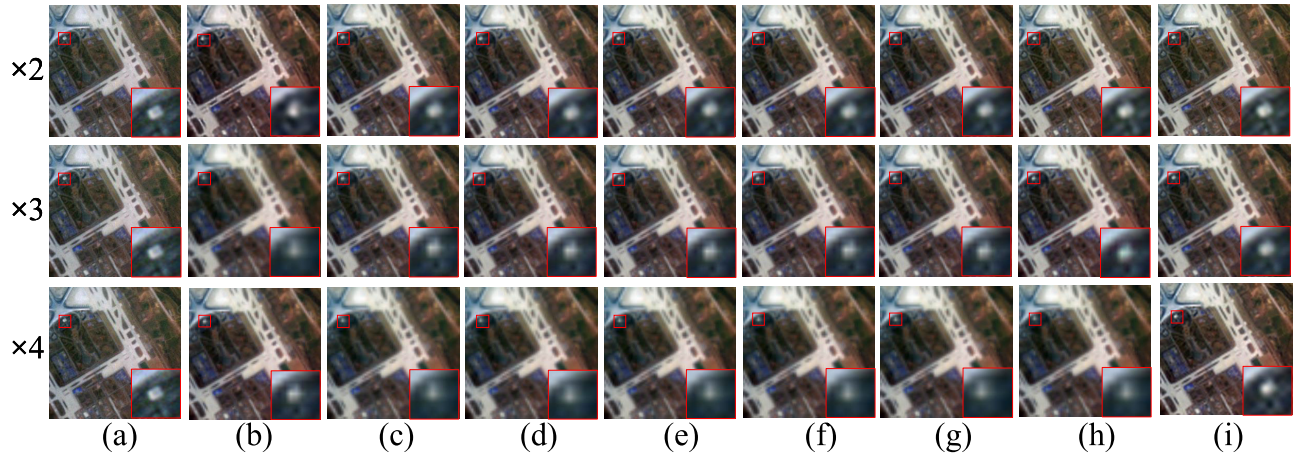


Fig. 11. Quantitative results of hyperspectral image super-resolution on the QUST-1 dataset. (a) Ground truth. (b) Bicubic. (c) MCNet [37]. (d) SSPSR [29]. (e) GELIN [30]. (f) EDiffSR [33]. (g) SNLSR [34]. (h) DBSR [35]. (i) Our framework.

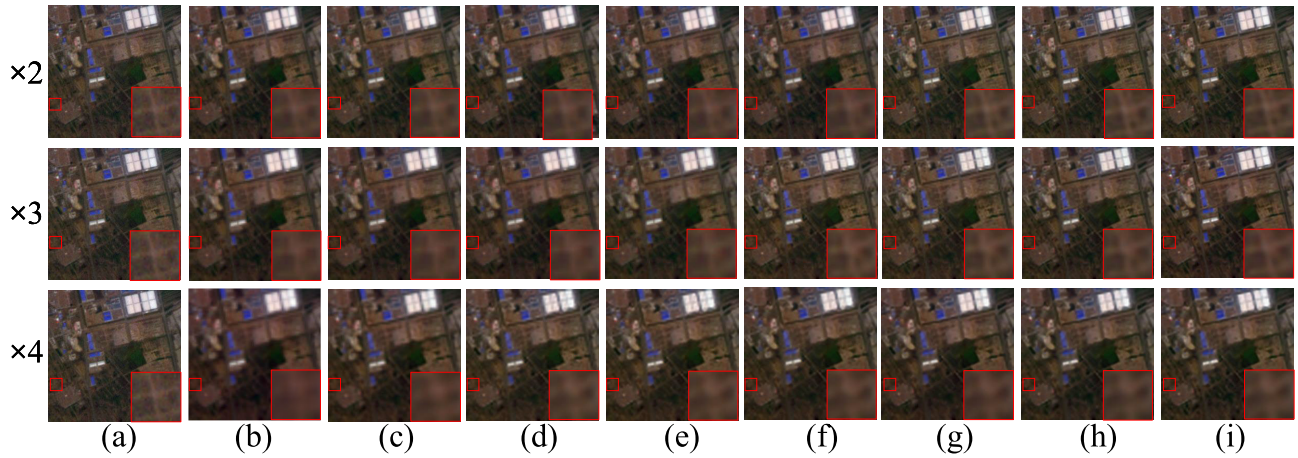


Fig. 12. Quantitative results of hyperspectral image super-resolution on the QUST-1 dataset. (a) Ground truth. (b) Bicubic. (c) MCNet [37]. (d) SSPSR [29]. (e) GELIN [30]. (f) EDiffSR [33]. (g) SNLSR [34]. (h) DBSR [35]. (i) Our framework.

Qingdao region dataset. The results are shown for the spectral bands 17-7-2 as R-G-B at scale factors 2, 3, and 4. The red rectangle in the images highlights regions with rich edge and texture details, presenting a challenge for hyperspectral image super-resolution. In the red rectangle of Figs. 11 and 12, methods like MCNet, SSPSR, GELIN, and SNLSR produce overly smooth contours, while EDiffSR results exhibit image distortions. In contrast, DBSR produces more visually realistic results. However, our Mamba framework captures extensive long-range pixel dependencies, generating images that closely resemble real-world objects. Figs. 13 and 14 show absolute error maps that highlight the differences between the reconstructed images and the ground truth. Smaller error values indicate better performance of the super-resolution methods. The images produced by our framework in Figs. 13 and 14 exhibit the smallest errors. This demonstrates its ability to accurately reconstruct spatial information. Fig. 15(b) presents the spectral curves of randomly selected pixel points (highlighted as red dots) in the reconstructed images derived from the QUST-1 dataset at a scale factor of 4. The spectral curves produced by our Mamba framework exhibit a high degree of alignment with the ground truth when compared to those generated by alternative methods. This demonstrates the advantage of our Mamba hyperspectral super-resolution framework in maintaining spectral accuracy, further validating the effectiveness of using the spectral feature dimension as the sequence axis for reconstructing global spectral information in hyperspectral images.

E. Complexity Evaluation

We evaluate the computational complexity and runtime of our framework with a $\times$ 4 upsampling factor on the QUST-1 dataset. We compare it to the other methods, as summarized in Table III. We assess the qualitative metrics, including parameter counts, FLOPs, and inference time [60]. The parameter and FLOPs metrics are relatively high. This is due to our framework's deep network architecture and the addition of spectral and spatial global effective receptive fields. Our framework has the fastest inference time compared to the other methods. This is because our framework significantly reduces the computational complexity of processing long sequences. This is achieved through the structured vision state-space model. The model optimizes information transmission efficiency and enables fast inference.

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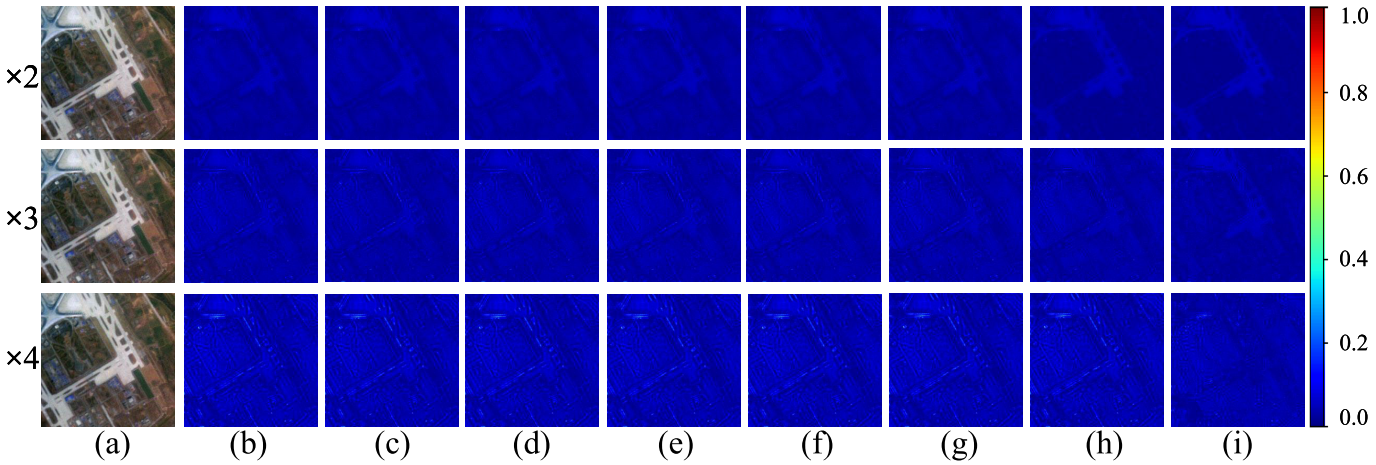


Fig. 13. Quantitative results of hyperspectral image super-resolution on the QUST-1 dataset. (a) Ground truth. (b) Bicubic. (c) MCNet [37]. (d) SSPSR [29]. (e) GELIN [30]. (f) EDiffSR [33]. (g) SNLSR [34]. (h) DBSR [35]. (i) Our framework.

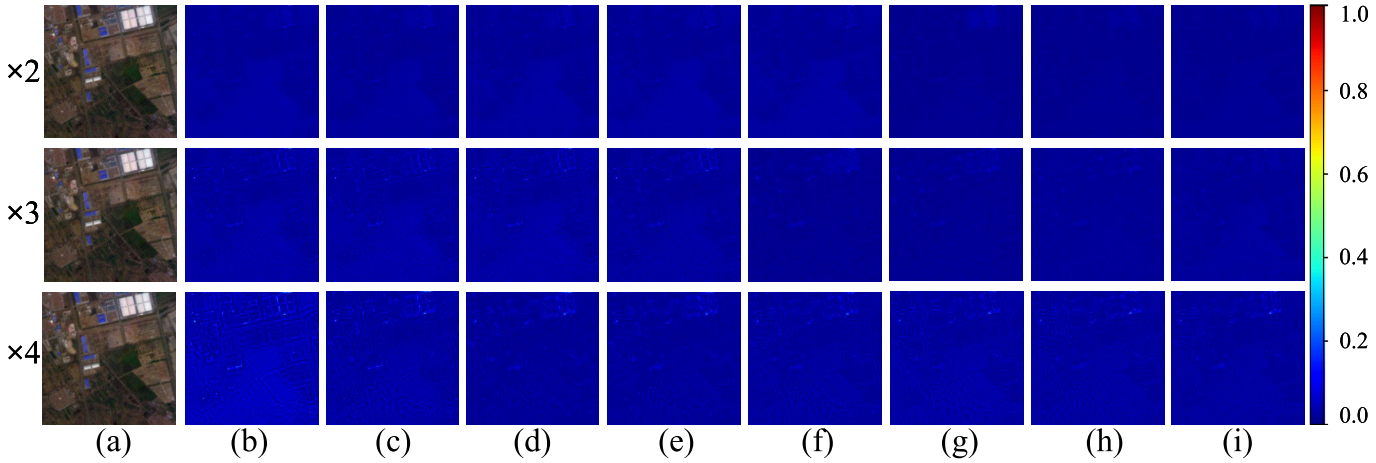


Fig. 14. Quantitative results of hyperspectral image super-resolution on the QUST-1 dataset. (a) Ground truth. (b) Bicubic. (c) MCNet [37]. (d) SSPSR [29]. (e) GELIN [30]. (f) EDiffSR [33]. (g) SNLSR [34]. (h) DBSR [35]. (i) Our framework.

F. Ablation Studies

In this section, to validate the effectiveness of each component of our framework, we conduct extensive ablation experiments on the QUST-1 dataset with an upsampling factor of 4. These experiments primarily focus on five aspects: 1) different designs of the spatial Mamba subnetwork; 2) different designs of the spectral Mamba subnetwork; 3) the impact of different scan modes on the overall performance of the framework; 4) different numbers of spatial Mamba blocks and spectral Mamba blocks; and 5) different designs of upsampling strategies.

1) Different Designs of Spatial Mamba Subnetwork: Table IV illustrates the influence of several designs of the spatial Mamba subnetwork on the QUST-1 dataset. "NSpaVSSM" represents that VSSM supersedes SpaVSSM inside our framework. "NMSFE" represents multiscale feature extraction with a corresponding number of convolutions. "NAGCAa" represents the omission of the AGCA mechanism inside the spatial Mamba subnetwork. "Our" represents the comprehensive Mamba hyperspectral image super-resolution framework. Table III indicates that our framework attains superior performance, while the variant using NSpaVSSM has the least favorable outcomes across all criteria. This highlights the necessity of fully utilizing spatial information and demonstrates the superior performance of the spatial Mamba subnetwork in this regard.

TABLE III

COMPARATIVE ANALYSIS OF DIFFERENT METHODS PARAMETERS, RUNNING TIME, AND COMPUTATIONAL COMPLEXITY

Method	Parameter (M)	FLOPs (G)	Inference time (s)
Bicubic	-	-	0.0060
MCNet	14.6485	5368.5677	0.3182
SSPSR	17.5620	6577.1888	0.4648
GELIN	15.5337	796.1502	1.0105
EDiffSR	5803.1036	204.3019	26.7160
SNLSR	1.4415	13.1793	0.2071
DBSR	407.4598	396.1162	0.1487
Our	1097.6826	111.9866	0.0268

2) Different Designs of Spectral Mamba Subnetwork: Table V illustrates the influence of several designs of the spectral Mamba subnetwork on the performance of the framework. "NSpeVSSM" represents the substitution of SpeVSSM with VSSM inside our framework. "NAGCAe" represents the absence of the AGCA mechanism inside the spectral Mamba

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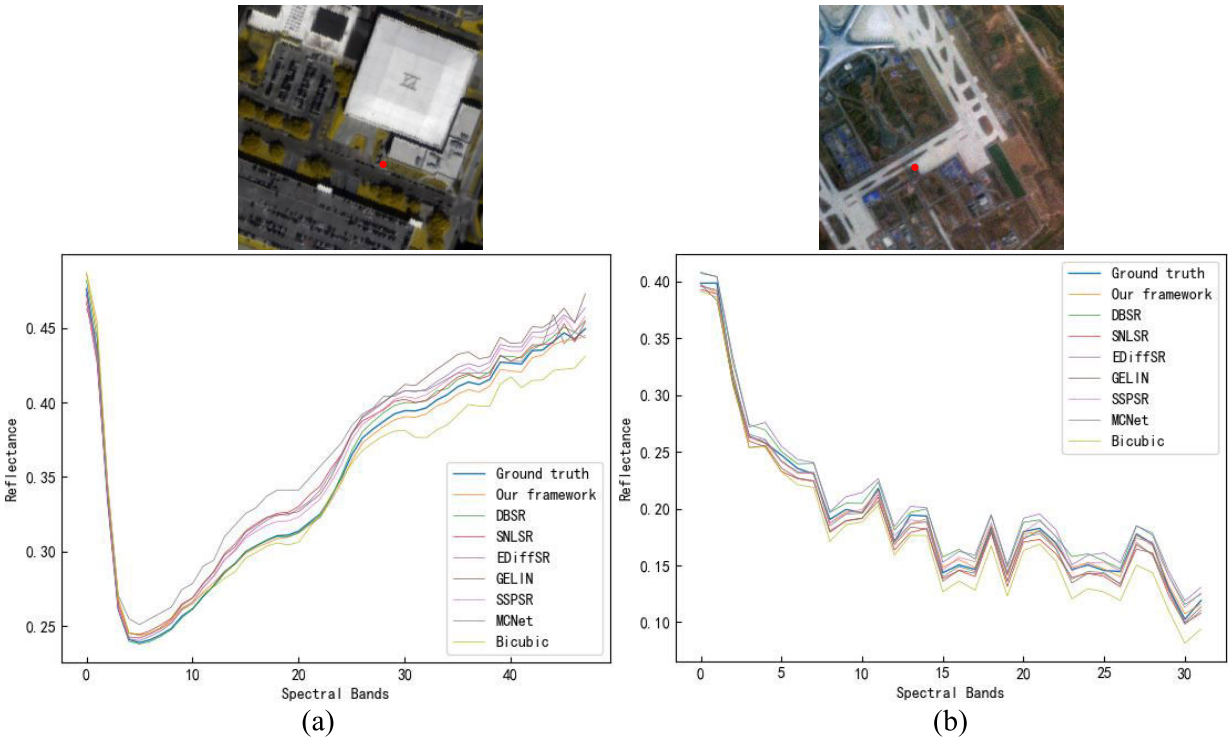


Fig. 15. Spectral curves of selected locations on (a) Houston and (b) QUST-1 datasets.

TABLE IV
RESULTS OF DIFFERENT DESIGNS OF SPATIAL MAMBA SUBNETWORK ON THE QUST-1 DATASET

	NSpaVSSM	NMSFE	NAGCAa	Our
FID ↓	75.3842	73.2345	72.8734	**71.3842**
PSNR ↑	36.2159	36.8497	37.4659	**37.8659**
SSIM ↑	0.8724	0.8916	0.9092	**0.9182**
SAM ↓	4.0532	3.8960	3.7852	**3.3591**
ERGAS ↓	2.1224	1.9753	1.8182	**1.7580**
CC ↑	0.8805	0.8918	0.9001	**0.9136**
LPIPS ↓	0.0362	0.0304	0.0281	**0.0266**
DISTS ↓	0.1597	0.1456	0.1347	**0.1283**

TABLE V
RESULTS OF DIFFERENT DESIGNS OF SPECTRAL MAMBA SUBNETWORK ON THE QUST-1 DATASET

	NSpeVSSM	NAGCAe	Our
FID ↓	72.8625	72.1186	**71.3842**
PSNR ↑	37.1397	37.5346	**37.8659**
SSIM ↑	0.9013	0.9105	**0.9182**
SAM ↓	4.1458	3.8532	**3.3591**
ERGAS ↓	1.8731	1.8007	**1.7580**
CC ↑	0.9021	0.9069	**0.9136**
LPIPS ↓	0.0311	0.0274	**0.0266**
DISTS ↓	0.1395	0.1317	**0.1283**

subnetwork. "Our" represents the comprehensive Mamba hyperspectral image super-resolution framework. Table V demonstrates that our architecture attains superior performance. This illustrates the significance of the spectral mamba subnetwork in identifying nuanced variations and patterns in global spectral data and in choosing essential spectral characteristics for precise spectral information reconstruction.

TABLE VI
RESULTS OF DIFFERENT SCAN MODES ON THE QUST-1 DATASET

	Spatial2	Spectral4	Our
FID ↓	71.9858	71.5327	**71.3842**
PSNR ↑	37.3261	37.4857	**37.8659**
SSIM ↑	0.9112	0.9135	**0.9182**
SAM ↓	3.7718	3.3613	**3.3591**
ERGAS ↓	1.8245	1.6674	**1.7580**
CC ↑	0.9014	0.9108	**0.9136**
LPIPS ↓	0.0295	0.0271	**0.0266**
DISTS ↓	0.1346	0.1309	**0.1283**

3) Different Scan Modes: Table VI illustrates the effects of various scan modes on the framework's performance. Within our framework (Our), the spatial Mamba subnetwork utilizes four separate directions for scanning to produce the scan sequence, while the spectral Mamba subnetwork uses bidirectional scanning (from the first band to the final band) to construct the scan sequence. Spatial2 represents the use of two orientations (top-left to bottom-right, bottom-right to top-left) for the scanning of spatial data to generate the scan sequence. Spectral4 represents the use of four distinct orientations for the acquisition of spectral data to produce the scanning sequence. Table VI indicates that our framework attains superior performance. In the spatial Mamba subnetwork, using four-directional spatial information scanning enables anchor pixels to perceive a larger neighborhood, thus enhancing the model's effectiveness. In the spectral Mamba subnetwork, since spectral information primarily consists of 1-D features, bidirectional scanning is sufficient to capture relevant information along the spectral dimension.

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TABLE VII

RESULTS OF DIFFERENT NUMBERS OF THE SPATIAL MAMBA BLOCKS AND THE SPECTRAL MAMBA BLOCKS ON THE QUST-1 DATASET

	SpaMB -14	SpaMB -15	SpatMB -16	SpeMB -2	SpeMB -3	SpeMB -4
FID↓	72.2332	**71.3842**	71.7864	71.9732	**71.3842**	71.6742
PSNR↑	37.6366	**37.8659**	37.5238	37.7074	**37.8659**	37.7435
SSIM↑	0.9104	**0.9182**	0.9163	0.9111	**0.918**2	0.9161
SAM↓	**3.3387**	3.3591	3.3986	3.8753	3.3591	3.3625
RGAS↓	1.8002	**1.7580**	1.9653	1.8643	**1.7580**	1.8357
CC↑	0.9121	0.9136	0.9019	0.9127	0.9136	**0.9157**
LPIPS↓	0.0276	**0.0266**	0.0283	0.0279	**0.0266**	0.0284
DISTS↓	0.1297	**0.1283**	0.1299	0.1288	**0.1283**	0.1295

TABLE VIII

RESULTS OF DIFFERENT DESIGNS OF THE UPSAMPLING STRATEGIES ON THE QUST-1 DATASET

	SingleUp	ProgresUp
FID ↓	73.2987	**71.3840**
PSNR ↑	37.30854	**37.8659**
SSIM ↑	0.91127	**0.9182**
SAM ↓	3.4219	**3.3591**
RGAS ↓	1.7975	**1.7580**
CC ↑	0.9011	**0.9136**
LPIPS ↓	0.0285	**0.0266**
DISTS ↓	0.1306	**0.1283**

4) Different Numbers of the Spatial Mamba Blocks and the Spectral Mamba Blocks: Table VII illustrates the effect of different numbers of spatial Mamba blocks and spectral Mamba blocks on the QUST-1 dataset. "SpaMB-" and "SpeMB-" represent the varying numbers of spatial and spectral Mamba blocks within our framework, respectively. As shown in Table VII, the spatial metric performs best with 15 spatial Mamba blocks, while the spectral metric performs best with three spectral Mamba blocks. Therefore, in the experiment, we set the number of spatial Mamba blocks to 15 and the number of spectral Mamba blocks to 3 to achieve the best quantitative metrics. The ablation study results indicate that our framework requires 15 spatial Mamba blocks to effectively capture spatial details due to their complexity and fine granularity. Using 15 spatial Mamba blocks significantly enhances the spatial fidelity of hyperspectral image super-resolution. In contrast, our framework efficiently extracts and processes essential spectral information with only three spectral Mamba blocks, preventing potential overfitting or redundancy in spectral feature extraction. Therefore, our framework balances the capture of detailed spatial structures with the preservation of robust spectral representations, leading to superior performance.

5) Different Designs of the Upsampling Strategies: Table VIII illustrates the effect of different designs of upsampling strategies on the QUST-1 dataset. "SingleUp" represents using single upsampling strategies in the outer layer of high-quality reconstruction. "ProgresUp" represents the upsampling strategies used in our framework. As shown in Table VII, ProgresUp outperforms the others in terms of metrics. This suggests that ProgresUp achieves better high-quality reconstruction by incrementally refining image details compared to SingleUp strategies.

VII. CONCLUSION

In this work, we are the first to explore the role of Mamba in hyperspectral image super-resolution, which addresses the problems of insufficient detail recovery and spectral information loss in high-dimensional hyperspectral images. Specifically, we designed the spatial Mamba model, which is used to capture a wide range of pixel spatial long-range dependencies. Capturing this dependency allows the network to access information from a broader area, maximizing the utilization of spatial information. We also designed a spectral Mamba subnetwork, which uses the spectral feature dimension as the sequence axis to capture subtle changes and patterns in the global spectral information, thereby achieving consistency in spectral super-resolution results. Finally, the reconstruction model gradually restores image details and spectral features through iterative upsampling, achieving high-quality reconstruction results. Our Mamba hyperspectral image super-resolution framework integrates these three parts, which can reconstruct hyperspectral images close to real objects while dealing with the complexity of large-scale and high-dimensional hyperspectral images. Meanwhile, the framework produces visually pleasing super-resolution images with rich spatial details. Extensive qualitative and quantitative evaluations confirm that our framework outperforms seven other hyperspectral image super-resolution methods. Future work will aim to improve the scalability of the framework. Additionally, further research will explore MambaHSISR's application to other remote sensing tasks, including object recognition, land change detection, and segmentation.

REFERENCES

[1] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[2] X. Sun, L. Zhuang, L. Gao, H. Gao, X. Sun, and B. Zhang, "Information entropy estimation based on point-set topology for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[3] X. Tao, M. E. Paoletti, Z. Wu, J. M. Haut, P. Ren, and A. Plaza, "An abundance-guided attention network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[4] C. Li et al., "CasFormer: Cascaded transformers for fusion-aware computational hyperspectral imaging," *Inf. Fusion*, vol. 108, Aug. 2024, Art. no. 102408.

[5] H. Wang, K. Köser, and P. Ren, "Large foundation model empowered discriminative underwater image enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[6] X. Li, W. Sun, Y. Ji, and W. Huang, "A joint detection and tracking paradigm based on reinforcement learning for compact HFSWR," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 1995–2009, 2025.

[7] W. Zhang, H. Wang, P. Ren, and W. Zhang, "Underwater image color correction via color channel transfer," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[8] Y. Fu, Z. Liu, and J. Lyu, "Transferable adversarial attacks for remote sensing object recognition via spatial-frequency co-transformation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[9] Y. Fu, Z. Liu, and J. Lyu, "Reason and discovery: A new paradigm for open set recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 63, pp. 1–14, Jan. 2025.

[10] P. Ren et al., "Oil spill drift prediction enhanced by correcting numerically forecasted sea surface dynamic fields with adversarial temporal convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[11] P. Ren, Z. Han, Z. Yu, and B. Zhang, "Confucius tri-learning: A paradigm of learning from both good examples and bad examples," *Pattern Recognit.*, vol. 163, Jul. 2025, Art. no. 111481.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:53:11 UTC from IEEE Xplore. Restrictions apply.

[12] H. Wang et al., "INSPIRATION: A reinforcement learning-based human visual perception-driven image enhancement paradigm for underwater scenes," *Eng. Appl. Artif. Intell.*, vol. 133, Jul. 2024, Art. no. 108411.

[13] H. Wang, S. Sun, and P. Ren, "Underwater color disparities: Cues for enhancing underwater images toward natural color consistencies," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 2, pp. 738–753, Feb. 2024.

[14] Q. Yan et al., "From dynamic to static: Stepwisely generate HDR image for ghost removal," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 2, pp. 1409–1421, Feb. 2025.

[15] Q. Yan et al., "Toward high-quality HDR deghosting with conditional diffusion models," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 4011–4026, May 2024.

[16] M. Wang, L. Gao, L. Ren, X. Sun, and J. Chanussot, "Hyperspectral simultaneous anomaly detection and denoising: Insights from integrative perspective," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13966–13980, 2024.

[17] F. Zhou, X. Sun, C. Sun, J. Dong, and X. X. Zhu, "Adaptive morphology filter: A lightweight module for deep hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[18] Y. Yang, X. Sun, J. Dong, K.-M. Lam, and X. Xiang Zhu, "Attention-ConvNet network for ocean-front prediction via remote sensing SST images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[19] Q. Zhu, M. Zhang, Y. Chen, G. Zheng, and J. Luo, "Spectral correlation-based fusion network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[20] G. Chen, J. Nie, W. Wei, L. Zhang, and Y. Zhang, "Arbitrary-scale hyperspectral image super-resolution from a fusion perspective with spatial priors," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[21] J. Liu, Z. Wu, and L. Xiao, "A spectral diffusion prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[22] Y. Xu et al., "Hyperspectral image super-resolution with ConvLSTM skip-connections," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[23] S. Weng et al., "Reference-based image super-resolution of hyperspectral and red-green-blue image for determination of wheat kernel quality using deep learning networks," *Eng. Appl. Artif. Intell.*, vol. 139, Jan. 2025, Art. no. 109513.

[24] S. Chen, L. Zhang, and L. Zhang, "Cross-scope spatial–spectral information aggregation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 5878–5891, 2024.

[25] S. Jiang, N. Li, M. Xu, S. Zhang, and S. Jia, "SQformer: Spectral-query transformer for hyperspectral image arbitrary-scale super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[26] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Apr. 2023, Art. no. 5509417.

[27] H. Yu, Z. Ling, K. Zheng, L. Gao, J. Li, and J. Chanussot, "Unsupervised hyperspectral and multispectral image fusion with deep spectral–spatial collaborative constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[28] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[29] J. Jiang, H. Sun, X. Liu, and J. Ma, "Learning spatial–spectral prior for super-resolution of hyperspectral imagery," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 1082–1096, 2020.

[30] X. Wang, Q. Hu, J. Jiang, and J. Ma, "A group-based embedding learning and integration network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541416.

[31] Y. Long, X. Wang, M. Xu, S. Zhang, S. Jiang, and S. Jia, "Dual self-attention Swin transformer for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[32] B. Wang, J. Chen, H. Wang, Y. Tang, J. Chen, and Y. Jiang, "A spectral and spatial transformer for hyperspectral remote sensing image super-resolution," *Int. J. Digit. Earth*, vol. 17, no. 1, Dec. 2024, Art. no. 2313102.

[33] Y. Xiao, Q. Yuan, K. Jiang, J. He, X. Jin, and L. Zhang, "EDiffSR: An efficient diffusion probabilistic model for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[34] Q. Hu, X. Wang, J. Jiang, X.-P. Zhang, and J. Ma, "Exploring the spectral prior for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 5260–5272, 2024.

[35] P. Yang, Y. Ma, X. Mei, Q. Chen, M. Wu, and J. Ma, "Deep blind super-resolution for hyperspectral images," *Pattern Recognit.*, vol. 157, Jan. 2025, Art. no. 110916.

[36] Q. Ma, J. Jiang, X. Liu, and J. Ma, "Learning a 3D-CNN and transformer prior for hyperspectral image super-resolution," *Inf. Fusion*, vol. 100, Dec. 2023, Art. no. 101907.

[37] Q. Li, Q. Wang, and X. Li, "Mixed 2D/3D convolutional network for hyperspectral image super-resolution," *Remote Sens.*, vol. 12, no. 10, p. 1660, May 2020.

[38] L. Bai, H. Li, W. Gao, J. Xie, and H. Wang, "A joint multiobjective optimization of feature selection and classifier design for high-dimensional data classification," *Inf. Sci.*, vol. 626, pp. 457–473, May 2023.

[39] K. Lagemann, C. Lagemann, B. Taschler, and S. Mukherjee, "Deep learning of causal structures in high dimensions under data limitations," *Nature Mach. Intell.*, vol. 5, no. 11, pp. 1306–1316, Oct. 2023.

[40] Z. Hu, Z. Shi, G. E. Karniadakis, and K. Kawaguchi, "Hutchinson trace estimation for high-dimensional and high-order physics-informed neural networks," *Comput. Methods Appl. Mech. Eng.*, vol. 424, May 2024, Art. no. 116883.

[41] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[42] T. Huang, X. Pei, S. You, F. Wang, C. Qian, and C. Xu, "LocalMamba: Visual state space model with windowed selective scan," 2024, *arXiv:2403.09338*.

[43] M. Ahmad et al., "Multi-head spatial–spectral mamba for hyperspectral image classification," *Remote Sens. Lett.*, vol. 16, no. 4, pp. 339–353, Apr. 2025.

[44] W. Li, F. Yuan, H. Zhang, Z. Lv, and B. Wu, "Hyperspectral object detection based on spatial–spectral fusion and visual mamba," *Remote Sens.*, vol. 16, no. 23, p. 4482, Nov. 2024.

[45] C. Cheng, H. Wang, and H. Sun, "Activating wider areas in image super-resolution," 2024, *arXiv:2403.08330*.

[46] H. Guo, J. Li, T. Dai, Z. Ouyang, X. Ren, and S. Xia, "MambaIR: A simple baseline for image restoration with state-space model," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Sep. 2024, pp. 222–241.

[47] G. Zhao, H. Wu, D. Luo, X. Ou, and Y. Zhang, "Spatial–spectral interaction super-resolution CNN–Mamba network for fusion of satellite hyperspectral and multispectral image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 18489–18501, 2024.

[48] J. Qiao et al., "Hi-mamba: Hierarchical mamba for efficient image super-resolution," 2024, *arXiv:2410.10140*.

[49] Y. Xiao, Q. Yuan, K. Jiang, Y. Chen, Q. Zhang, and C.-W. Lin, "Frequency-assisted mamba for remote sensing image super-resolution," 2024, *arXiv:2405.04964*.

[50] Z. Li, F. Liu, W. Yang, S. Peng, and J. Zhou, "A survey of convolutional neural networks: Analysis, applications, and prospects," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 6999–7019, Dec. 2021.

[51] X. Xiang, Z. Wang, J. Zhang, Y. Xia, P. Chen, and B. Wang, "AGCA: An adaptive graph channel attention module for steel surface defect detection," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.

[52] N. B. Bynagari, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," *Asian J. Appl. Sci. Eng.*, vol. 8, no. 1, pp. 25–34, Apr. 2019.

[53] L. C. Chan and P. Whiteman, "Hardware-constrained hybrid coding of video imagery," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-19, no. 1, pp. 71–84, Jan. 1983.

[54] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[55] F. A. Kruse et al., "The spectral image processing system (SIPS)-interactive visualization and analysis of imaging spectrometer data," *Remote Sens. Environ.*, vol. 44, nos. 2–3, pp. 145–163, May 1993.

[56] L. Wald, "Quality of high resolution synthesised images: Is there a simple criterion?" in *Proc. 3rd Conf. Fusion Earth Data, Merging Point Meas., Raster Maps Remotely Sensed Images*, 2000, pp. 99–103.

[57] L. Loncan et al., "Hyperspectral pansharpening: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[58] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 586–595.

[59] K. Ding, K. Ma, S. Wang, and E. P. Simoncelli, "Image quality assessment: Unifying structure and texture similarity," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2567–2581, May 2022.

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[60] W. Förstner, "Reliability analysis of parameter estimation in linear models with applications to mensuration problems in computer vision," *Comput. Vis., Graph., Image Process.*, vol. 39, no. 3, p. 392, Sep. 1987.



Yinghao Xu received the B.S. degree in Internet of Things engineering from Qilu University of Technology, Jinan, China, in 2020, and the M.S. degree in computer technology from Qingdao University of Science and Technology, Qingdao, China, in 2023. He is currently pursuing the Ph.D. degree in ocean resources and information engineering with the School of College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao.

His research interests include deep learning and reinforcement learning, as well as applications in remote sensing image processing.



Hao Wang received the B.S. and M.S. degrees in control engineering from Qingdao University of Technology, Qingdao, China, in 2017 and 2020, respectively. He is currently pursuing the Ph.D. degree in control science and engineering with the College of Control Science and Engineering, China University of Petroleum (East China), Qingdao.

His research interests include deep learning and reinforcement learning, with applications to underwater image processing.



Fei Zhou received the B.S., M.S., and Ph.D. degrees in electronic science and technology from the Ocean University of China, Qingdao, China, in 2016, 2019, and 2024, respectively.

He is currently a Lecturer with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao. His research interests include machine vision and pattern recognition.



Chunbo Luo (Senior Member, IEEE) is currently an Associate Professor of computer science at the University of Exeter, Exeter, U.K. His research has been generously supported by the Research Councils U.K., the Royal Society, the EU H2020, and various industries. His research interests focus on machine learning and engineering methods for remote sensing and Earth observations.

Dr. Luo is a fellow of the Higher Education Academy.



Xin Sun (Senior Member, IEEE) received the Ph.D. degree from the College of Computer Science and Technology, Jilin University, Changchun, China, in 2013.

He is a Lecturer with the Faculty of Data Science, City University of Macau, Macau SAR, China. His research interests include machine learning, data mining, and computer vision.



Susanto Rahardja (Fellow, IEEE) is currently a Professor of engineering cluster with Singapore Institute of Technology, Singapore, and the Ph.D. Advisor with Northwestern Polytechnical University, Xi'an, China. He contributed to the development of a series of audio compression technologies, such as Audio Video Standards AVS-L, AVS-2, ISO/IEC 14496-3:2005/Amd.2:2006, and ISO/IEC 14496-3:2005/Amd.3:2006, which have been licensed worldwide. His research interests include multimedia coding and processing, wireless communications, discrete transforms, machine learning, signal processing algorithms, implementation, and optimization. He has more than 15 years of experience in leading a research team in the above-mentioned areas.

Dr. Rahardja is a fellow of the Academy Engineering, Singapore. He was the past Associate Editor of IEEE TRANSACTIONS ON AUDIO, SPEECH, AND LANGUAGE PROCESSING and the past Senior Editor of IEEE JOURNAL OF SELECTED TOPICS IN SIGNAL PROCESSING. He is currently an Associate Editor of *Journal of Visual Communication and Image Representation* (Elsevier), IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS, and IEEE TRANSACTIONS ON MULTIMEDIA. He is an Editorial Board Member of IEEE ACCESS.



Peng Ren (Senior Member, IEEE) received the B.S. and M.S. degrees in electronic engineering from Harbin Institute of Technology, Harbin, China, in 2003 and 2005, respectively, and the Ph.D. degree in computer science from the University of York, York, U.K., in 2010.

He is currently a Professor with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao, China. His research interests include marine information processing, remote sensing, and machine learning.

Dr. Ren was a recipient of the K. M. Scott Prize from the University of York in 2011 and the Eduardo Caianiello Best Student Paper Award at the 18th International Conference on Image Analysis and Processing in 2015 as a coauthor. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

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